

Methodology

Written so an economist can attack it. Every choice below is a choice, and the ones that are arbitrary are labelled arbitrary.

1. The calculation

For policy p and household h :

$$\text{effect}(h, p) = \text{net}(h, p) - \text{net}(h, \text{baseline})$$

where `net` is annual household net income after federal income tax, state income tax, payroll tax and means-tested benefits, computed by PolicyEngine US for the simulation year.

Only the parameters listed on each policy's page are altered. Everything else in the tax and benefit system is held at current law. That is what makes the difference attributable to the policy and nothing else.

This is a static, first-order statutory calculation. No behavioural response, no labour supply elasticity, no general equilibrium, no macroeconomic feedback. A household that would work more hours under a larger EITC is modelled as working the same hours. This is a real limitation and it is the reason no figure here claims to forecast your finances.

2. The grid and interpolation

The rules engine is Python; the site is static. So the engine runs at build time over a grid of generic household shapes and the browser interpolates.

- States: CA, VA, TX, NY, FL, OH, PA, WA
- Filing status: single, married filing jointly (modelled as a single-earner couple)
- Children: 0–3, aged 4, 9 and 14 as household size grows
- Adults: aged 35
- Income: a non-uniform ladder, \$2,500 steps to \$50,000, then \$5,000 to \$100,000, then \$20,000 to \$300,000, then \$50,000 to \$600,000

The ladder is dense at the bottom because credit phase-ins and phase-outs are dense

there, and sparse at the top because the schedules are flat.

Tax and benefit schedules are piecewise linear between kinks, so linear interpolation is **exact** within any step that contains no kink, and slightly smoothed where a step contains one. `interpolationSlack` measures the curvature across neighbouring grid points and lowers confidence where it is large relative to the effect.

3. Implementation priors

Expected value = estimated effect × probability the policy is actually in force affecting you.

Stage	Prior
PROMISE	0.15
PROPOSED	0.25
PASSED	0.60
ENACTED	0.90
IMPLEMENTED	1.00
HISTORICAL	0.35

These are stated priors, not measurements. They are not derived from base rates of bill passage, and they should not be read as if they were. They exist to stop a campaign promise outranking an enacted law by claiming a bigger headline number. They are published here and in the interface precisely so that a reader who disagrees with a figure can recompute the expected value from the estimate.

`HISTORICAL` sits low deliberately: a policy having operated before tells you the mechanism and parameters are known, which raises *confidence in the size*, while telling you nothing reassuring about whether it will return, which is what the prior measures. Those two things are scored separately on purpose.

4. Confidence

Starts at 0.5, then moves on four things:

Factor	Effect
Evidence stage	−0.25 (promise) to +0.35 (in force)

Computed by a rules engine rather than by hand	+0.10
Income near a phase-in or phase-out edge	-0.15
Two or more stated modelling assumptions	-0.05
Incomplete profile	-0.10

Clamped to [0.05, 0.98]. High ≥ 0.72 , medium ≥ 0.45 , low below.

The score is not the output; the *reasons* are. Every policy page lists them in plain sentences, because a reader can argue with “your income sits near a phase-out edge” and cannot argue with “0.61”.

5. Reported ranges

The band widens as confidence falls: $\pm 8\%$ at high, $\pm 20\%$ at medium, $\pm 45\%$ at low. Rounding steps coarsen with magnitude, so a figure over \$2,000 is rounded to \$50.

This is a presentation rule, not a statistical interval. It exists to stop the interface implying that a number derived from a promise deserves the same number of significant figures as one derived from statute. \$1,743.26 is never displayed; \$1,750 or \$1,500–\$2,000 is.

6. Double counting

Every policy is measured against the same baseline, so two policies touching the same statutory quantity overlap and their figures **cannot be added**.

Each component carries a `mechanism tag`. `overlapWarnings` detects collisions and the dashboard refuses to display a combined total for them, saying why instead. A genuine combined figure requires simulating the policies jointly in the pipeline. The browser cannot fake that, and the interface does not pretend otherwise.

7. Priority ranking

The “what moves your money” list is annual dollars of exposure per category, computed from your profile: total tax burden from the engine, housing cost $\times 12$, benefits from the engine, transport $\times 12$, student loan interest, and an employment term that dominates when you are out of work. It is a magnitude ranking, not a preference ranking — it tells you which areas *can* move your finances most, not which you care about.

8. Attribution

Nothing in this build attributes an economic outcome to a named politician. When candidate data exists, the distinction between “this happened during their tenure” and “they caused this” must be made explicitly, and any figure of the first kind must not be scored as if it were the second.

9. Known limitations

- Static first-order effects only (§1)
- Housing cost is collected but does not feed the benefit calculation, so SNAP is understated for renters with high shelter costs
- All income modelled as W-2 wages from one earner; no self-employment, investment income, or capital gains
- Married filing separately, head of household and surviving spouse are not in the grid
- No local or municipal taxes
- No candidates, ballot measures or elections
- 8 states of 50
- Interpolation smooths kinks within a step

10. What would change these numbers

Upstream statutory changes encoded in PolicyEngine. The scheduled workflow reruns the pipeline monthly and opens a pull request when the output differs, so a human reviews the diff before the published figures move. CI independently verifies that the committed grid reproduces exactly from a fresh run, which is what stops anyone editing the numbers by hand.